ENGINEERING
McMaster-Mohawk
Bachelor of Technology
Partnership

McMaster University

Course Outline

1. COURSE INFORMATION

Session Offered	Fall 2026	
Course Name	Smart Cities and Communities	
Course Code	SMRTTECH 4SC3, SEP 6ES3	
Date(s) and Time(s) of lectures	C01 – Fridays 9:30 am – 12:20 pm (ETB 529) L01 - Fridays 12:30 pm - 3:20 pm (ETB 537) Sep 25, Oct 9, Oct 23, Nov 6, Nov 20, Dec 4 L02 - Wednesdays 6:30 pm - 9:20 pm (ETB 537) Sep 23, Oct 7, Oct 21, Nov 4, Nov 18, Dec 2	
Program Name	Automation Engineering Technology	
Calendar Description	Emerging technologies shaping smart cities and communities.	
Instructor(s)	Dr. Marjan Alavi, P.Eng. Lab instructor: Dr. Ahmad Mohammadi	alavis2@mcmaster.ca Office hours: Thursdays 1:30-3:30 ETB 507 mohaa318@mcmaster.ca

2. COURSE SPECIFICS

Course Description	This course covers the deployment of city and community networks, the computation of data extracted from connected devices in the city and the sharing of analyzed data across agencies. The course also covers city-wide system monitoring and predictive modelling to optimize and improve services such as parking and public transportation, low energy consumption, increase safety, reduce traffic congestion, and protect infrastructure.		
Instruction Type	**Code**	**Type**	**Hours per term**
	C	Classroom instruction	39
	L	Laboratory, workshop or fieldwork	18
	T	Tutorial	
	DE	Distance education	
		Total Hours	57
Resources	**ISBN**	**Textbook Title & Edition**	**Author & Publisher**

	ISBN: 9780429434983	Data Analytics for Smart Cities (1st ed.). Auerbach Publications. https://doi-org.libaccess.lib.mcmaster.ca/10.1201/9780429434983	Alavi, A., & Buttlar, W.G. (Eds.). (2018).
	Other Supplies	**Source**	

Prerequisite(s)	SMRTTECH 4ES3, 4ID3, ENGTECH 4EE0, and registration in level IV of Automation Engineering Technology
Corequisite(s)	
Antirequisite(s)	
Course Specific Policies	All lectures and labs are in person. Your active participation in the lab and lectures is required to gain quiz, lab, and other class activity grades. In case of absence, please communicate with your instructor before missing a class or as soon as possible and apply for MSAF for missed academic work where applicable.
Departmental Policies	Students must maintain a GPA of 3.5/12 to continue in the program. In order to achieve the required learning objectives, on average, B.Tech. students can expect to do at least 3 hours of "out-of-class" work for every scheduled hour in class. "Out-of-class" work includes reading, research, assignments and preparation for tests and examinations. Where group work is indicated in the course outline, such collaborative work is mandatory. The use of cell phones, iPods, laptops and other personal electronic devices are prohibited from the classroom during the class time, unless the instructor makes an explicit exception. Announcements made in class or placed on Avenue are considered to have been communicated to all students including those individuals that are not in class. Instructor has the right to submit work to software to identify plagiarism.

3. SUB TOPIC(S)		
Week 1 Sep 11	Course Introduction Group Formation	
Week 2 Sep 18	Roadway Traffic Simulation and its Application in Smart Cities	
Week 3 Sep 25	Intelligent Transportation Systems in Smart Cities: Signals & AVs	

Week 4 Oct 2	Graph Theory	
Week 5 Oct 9	Digital Twin in Smart Cities I	
Week 6 Oct 16	***Reading Week - No Class***	
Week 7 Oct 23	Digital Twin in Smart Cities II	
Week 8 Oct 30	Smart Railway Systems (Train Scheduling and Operations)	
Week 9 Nov 6	Smart Systems (Smart Parking)	
Week 10 Nov 13	**Miditerm Test**	
Week 11 Nov 20	Guest Lecture Demonstration of Web Prototyping	
Week 12 Nov 27	Data Challenges in Smart Cities (Existing)	
Week 13 Dec 4	**Final Presentation**	

Classes end: December 10
Final examination period: December 11 to Wednesday December 23
All examinations MUST be written during the scheduled examination period

List of experiments	
Lab 1	**Lab 1 (**Roadway Traffic Simulation and its Application in Smart Cities)
Lab 2	**Lab 2 (**ITS: Signals & AVs)
Lab 3	**Lab 3 (**Digital Twin I)
Lab 4	**Lab 4 (**Digital Twin II)
Lab 5	**Lab 5 (**Train Scheduling and Operations)
Lab 6	**Lab 6 (Dec 02&04)**

Note that this structure represents a plan and is subject to adjustment term by term.
The instructor and the University reserve the right to modify elements of the course during the term. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

4. ASSESSMENT OF LEARNING *including dates*	Weight
Assignments, Quizzes, and class activities (10)	20%
Mid-term test – (in-person)	20%
LabS	20%
Mini Project A	15%
Mini Project B	15%
Final Presentation	10%
TOTAL	**100%**
Percentage grades will be converted to letter grades and grade points per the University calendar.	

5. LEARNING OUTCOMES

1. Explain smart cities' characterizations, challenges and state-of-the-art solutions.
2. Describe smart city and community networks.
3. Apply artificial intelligence to optimize and improve city and community services.
4. Analyse and visualize data extracted from connected devices and communities.
5. Use cloud services to deploy Machine learning predictive models and NLP algorithms.
6. Propose and review city-wide system monitoring applications.
7. Practice teamwork and transferable skills.

6. COURSE OUTLINE – APPROVED ADVISORY STATEMENTS

ANTI-DISCRIMINATION

The Faculty of Engineering is concerned with ensuring an environment that is free of all discrimination. If there is a problem, individuals are reminded that they should contact the Department Chair, the Sexual Harassment Officer or the Human Rights Consultant, as soon as possible.
http://www.mcmaster.ca/policy/General/HR/Discrimination_Harassment_Sexual_Harassment-Prevention&Response.pdf

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/

The following illustrates only three forms of academic dishonesty: The following illustrate only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com, please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

COMMUNICATIONS

It is the student's responsibility to:

- Maintain current contact information with the University, including address, phone numbers, and emergency contact information.
- Use the University provided e-mail address or maintain a valid forwarding e-mail address.
- Regularly check the official University communications channels. Official University communications are considered received if sent by postal mail, fax, or e-mail to the student's designated primary e-mail account via their @mcmaster.ca alias.
- Accept that forwarded e-mails may be lost and that e-mail is considered received if sent via the student's @mcmaster.ca alias.
- Check the McMaster/Avenue email and course websites on a regular basis during the term.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests. http://www.mcmaster.ca/policy/Students-AcademicStudies/Studentcode.pdf

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.